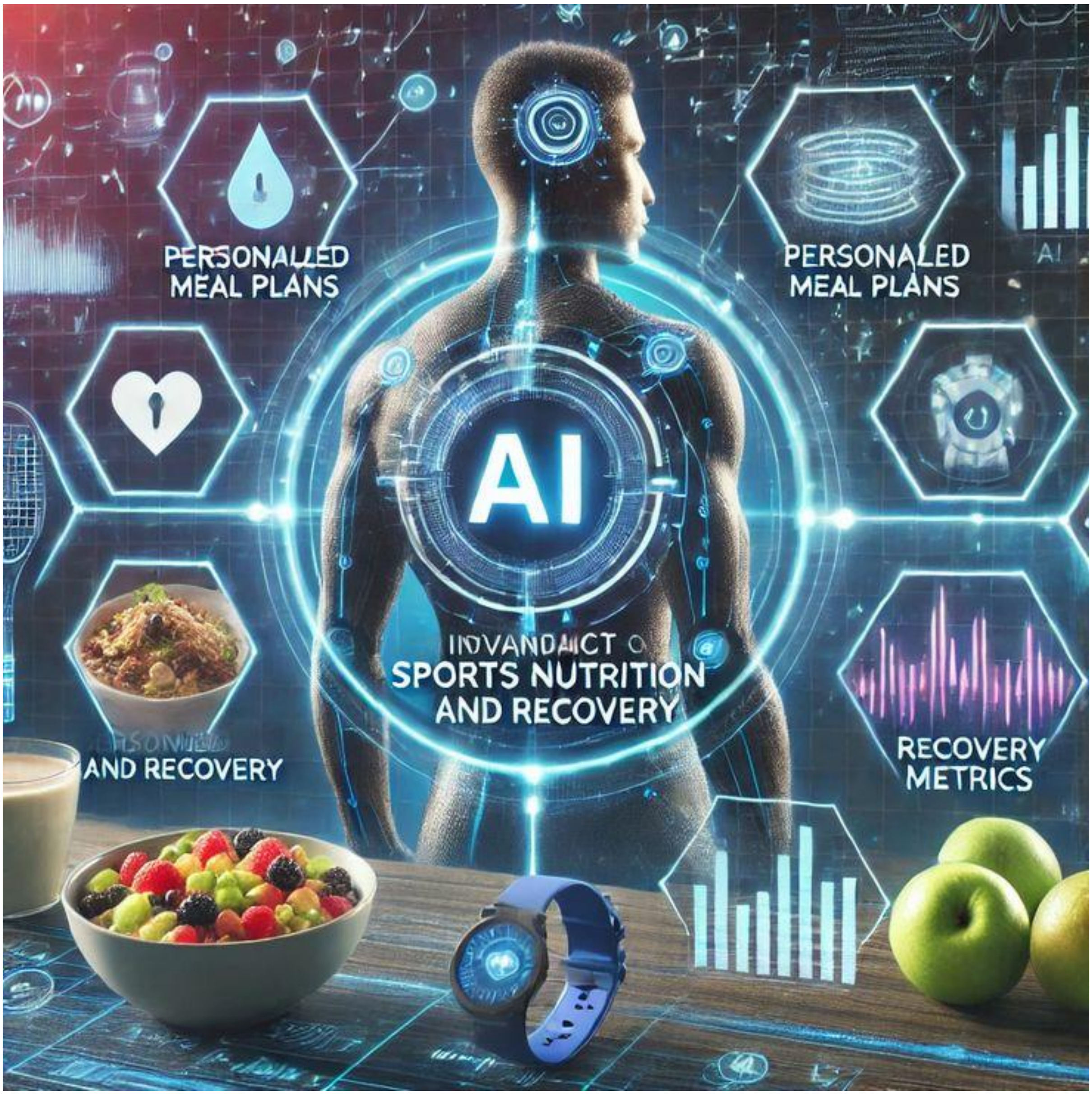

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AI-Based Food Recognition and Nutritional Analysis Model

Introduction

In recent years, the integration of **Artificial Intelligence (AI)** in everyday applications has significantly improved how we interact with technology. One such practical application is food recognition using image processing. This project focuses on developing an AI-based model that takes an image of food as input and accurately identifies the dish. Beyond simple classification, the model also provides detailed nutritional insights, including the proportion of carbohydrates, proteins, fats, and total calories.

This system aims to bridge the gap between technology and health awareness by enabling users to make informed dietary choices through a simple image input. With the help of machine learning and deep learning techniques, the model ensures high accuracy and user-friendly interaction through a graphical interface.

Objectives

- To develop an AI model capable of identifying food items from images
- To estimate nutritional values such as carbohydrates, proteins, fats, and calories
- To visualize nutritional information using graphical representations
- To achieve high accuracy in food classification using a standard dataset
- To deploy the model using an interactive web interface
- To promote awareness of healthy eating habits through technology

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Dataset Description – Food-101 (Kaggle)

The model is trained using the **Food-101 dataset**, available on Kaggle. This dataset is widely used for food image classification tasks.

Key Features of the Dataset:

- Contains **101 different food categories**
- Each category includes **1,000 images**
- Total of **101,000 images**
- Divided into:
    - **Training set**: 75,750 images

    - **Test set**: 25,250 images
- Images are real-world, noisy, and diverse, making the dataset suitable for robust model training

The dataset includes popular dishes such as pizza, burgers, sushi, pasta, desserts, and more, providing a broad base for classification.

Tools and Libraries Used

Development Tools:

- **Anaconda** – Environment and package management
- **VS Code** – Code development and debugging
- **Python** – Core programming language

Libraries:

- **Pandas** – Data handling and preprocessing
- **NumPy** – Numerical computations
- **Scikit-learn** – Model evaluation and preprocessing utilities
- **Matplotlib** – Data visualization and graph plotting
- **Streamlit** – Web application deployment
- **Joblib** – Model saving and loading

Methodology

1. Data Collection

- The Food-101 dataset is downloaded from Kaggle
- Images are organized into training and testing directories

2. Data Preprocessing

- Image resizing and normalization
- Label encoding for classification
- Splitting into training and validation sets

3. Model Development

- A deep learning-based image classification model is trained

- Feature extraction techniques are applied
- The model learns to identify patterns in food images

4. Model Training

- Training is performed using the prepared dataset
- Hyperparameters are tuned for optimal performance
- Training is accelerated using GPU (if available)

5. Model Evaluation

- Accuracy and loss metrics are calculated
- Performance is validated using the test dataset
- Confusion matrix and classification reports are generated

6. Model Deployment

- The trained model is saved using Joblib
- A user interface is created using Streamlit
- Users can upload images and receive predictions instantly

7. Results and Visualization

- The model outputs:
    - Food name (prediction)
    - Nutritional values (carbs, fats, proteins, calories)
- Graphs are generated using Matplotlib to visually represent nutritional proportions

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Results and Accuracy

The model demonstrates high accuracy in identifying food images due to the diversity and size of the Food-101 dataset. The integration of visualization tools enhances user understanding by presenting nutritional data in graphical form.

Future Scope

- Integration with real-time camera input for instant recognition
- Expansion of dataset to include more regional and homemade foods

- Improved nutritional estimation using external APIs
- Mobile application development for wider accessibility
- Incorporation of personalized diet recommendations

Importance of the Model

- Helps users make informed dietary decisions
- Promotes health awareness through technology
- Useful for fitness enthusiasts and diet planners
- Demonstrates practical application of AI in daily life

Conclusion

This project successfully demonstrates the application of AI in food recognition and nutritional analysis. By combining image classification with data visualization, the model provides both accuracy and usability. It highlights the potential of AI in improving health awareness and simplifying complex tasks through intuitive solutions.

Author

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References

- Kaggle – Food-101 Dataset
- Scikit-learn Documentation
- Streamlit Documentation
- NumPy and Pandas Official Documentation